

ASTR 302 Fall 2026

Coordinates and the Celestial Sphere



**Sloan Foundation
Telescope
New Mexico, U.S.A.**



**du Pont Telescope
Chile**



Agenda – August 31, 2026

- Class Logistics
- Bright Object Limit Revisit
- The Importance of Coordinates, Distances, and Astrometry
- How Coordinates are Defined on the Celestial Sphere
- How and why certain astronomical objects move over time

Recommended Reading: Birney Chapter 1 and 4

Class Logistics

- Dr. Miles may be OUT Wednesday. **BUT THERE WILL BE CONTENT.**
- Reach out to your point of contact/TAs for your final project to discuss feasibility.
- Observing Proposals (**individual**) are Due Friday at 11:59 pm (**9/11**) on D2L

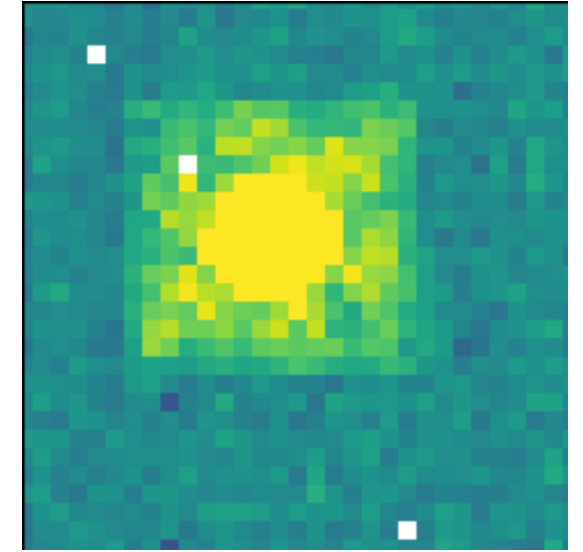
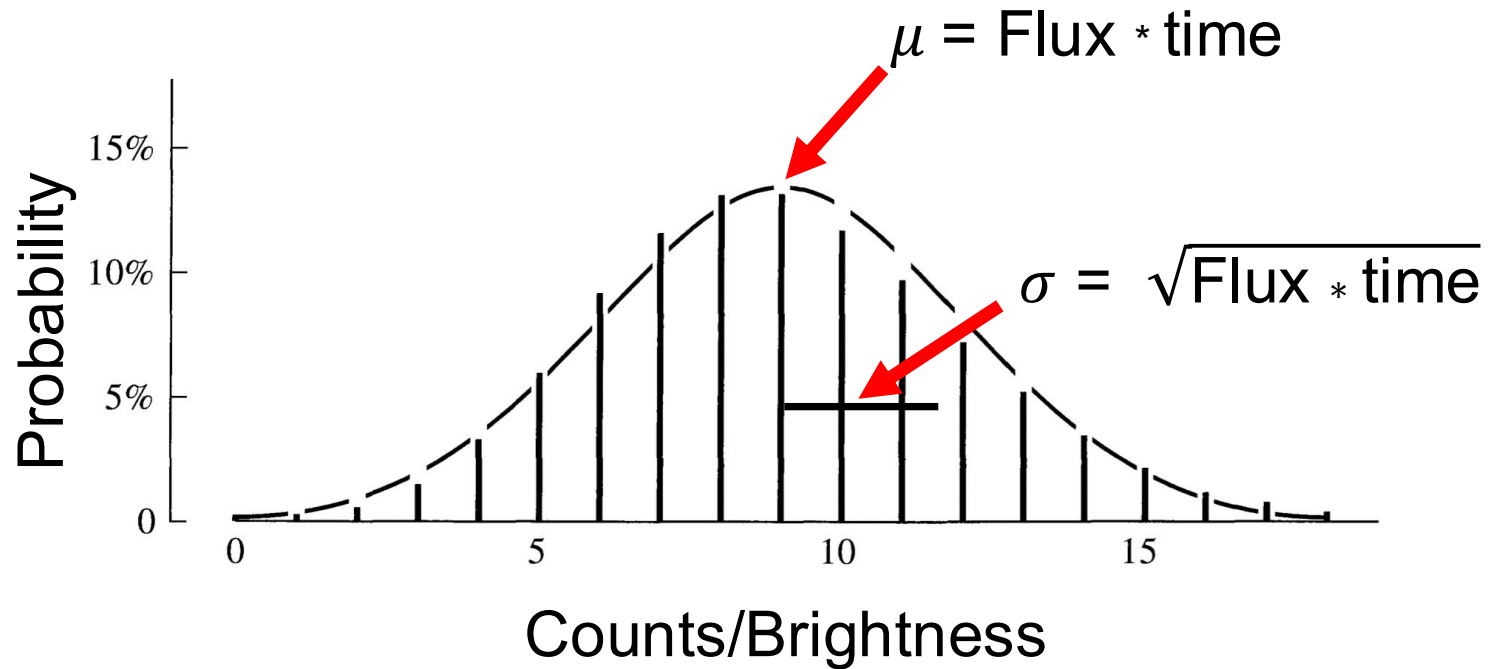
Looking Ahead

- 5 – minute presentations + questions describing your project (**Monday 9/21**)
(**5% of your Final Project Grade**)

Random Noise from Photons

- Random noise from astronomical sources is often caused by the random behavior of **photons**.
- Photons are discrete particles that are emitted or reflected from a source. Either 0, 1, 2.. etc photons
- The Poisson distribution describes the probability of observing N events over a fixed interval of time assuming some fixed event rate.
- For a sufficiently high event rate, the Poisson distribution behaves like a Gaussian distribution.
- **You rarely directly measure flux, just events or counts which are unitless**

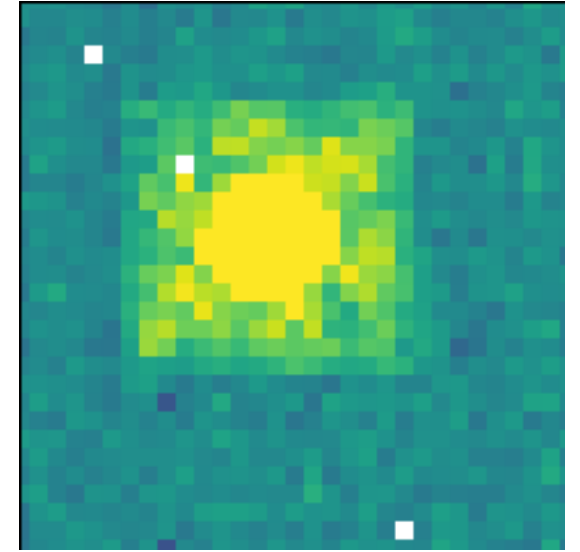
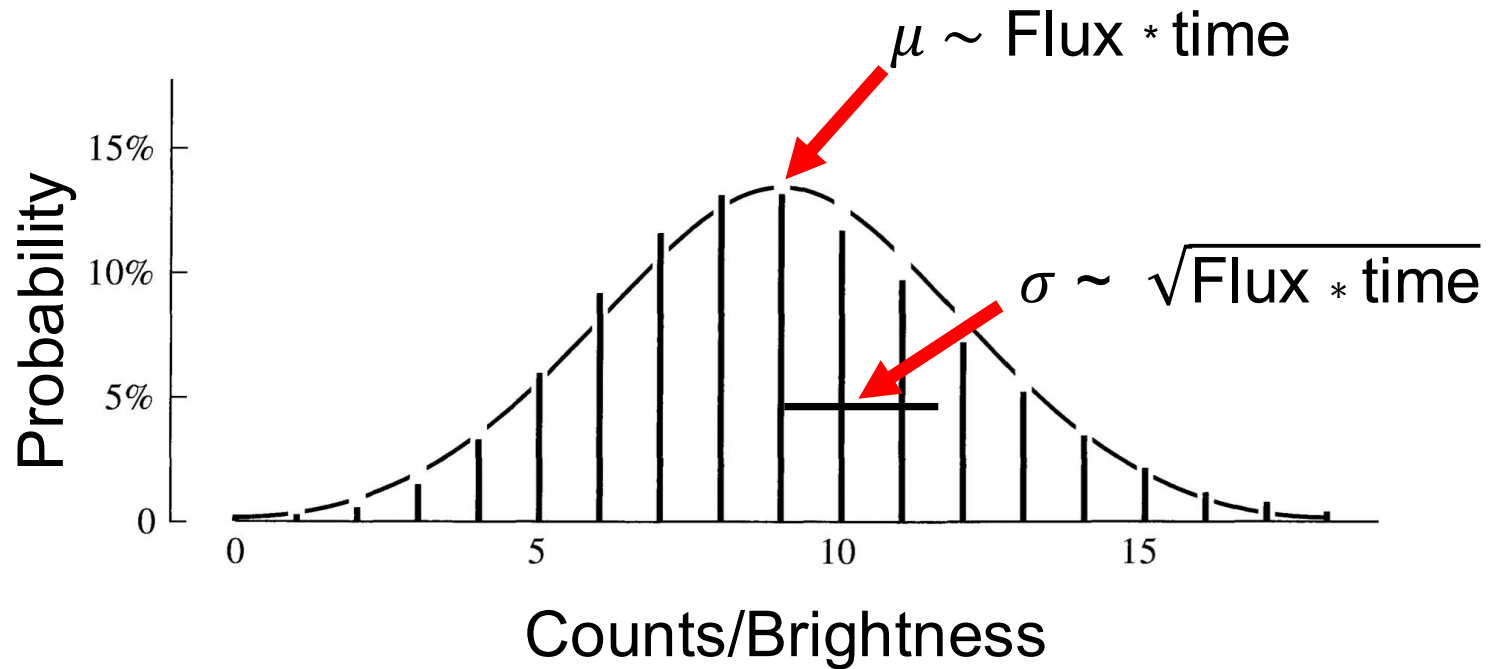
Photon Noise Limit Application: Bright Objects



Expected Value or Mean (μ) = Flux * time

Standard Deviation or Error (σ) = $\sqrt{\text{Flux} * \text{time}}$

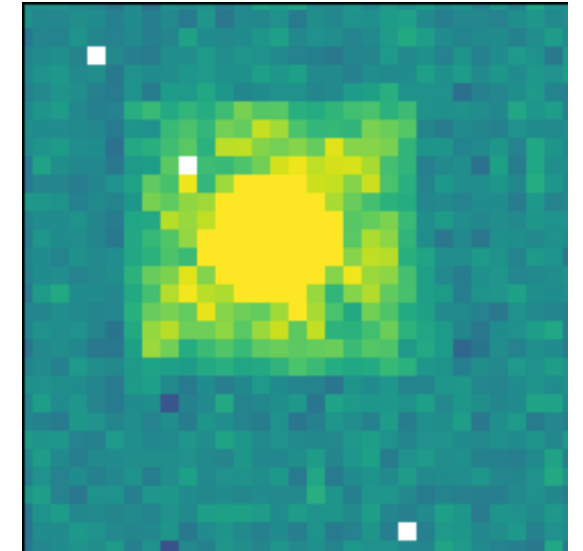
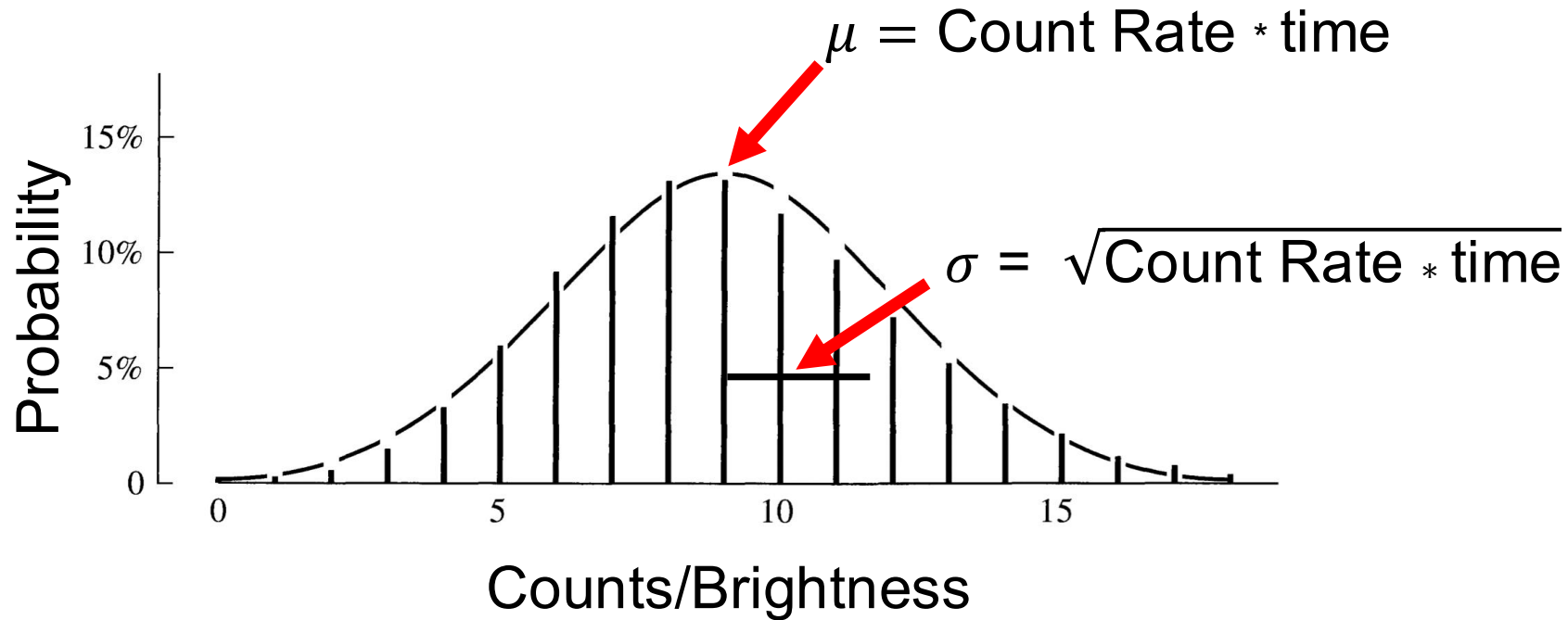
Photon Noise Limit Application: Bright Objects



Expected Value or Mean (μ) $\sim \text{Flux} * \text{time}$

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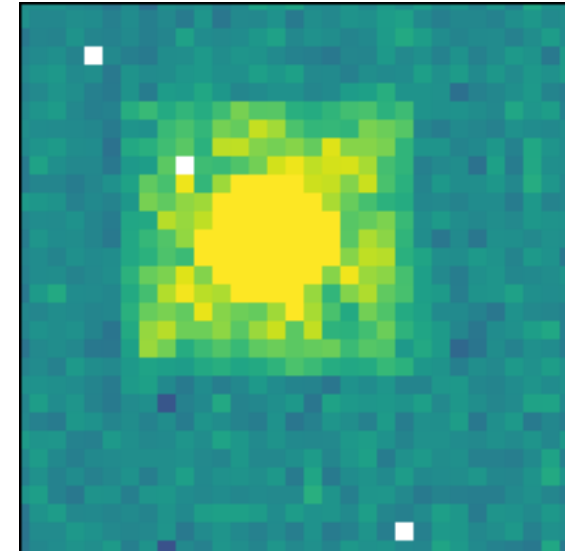
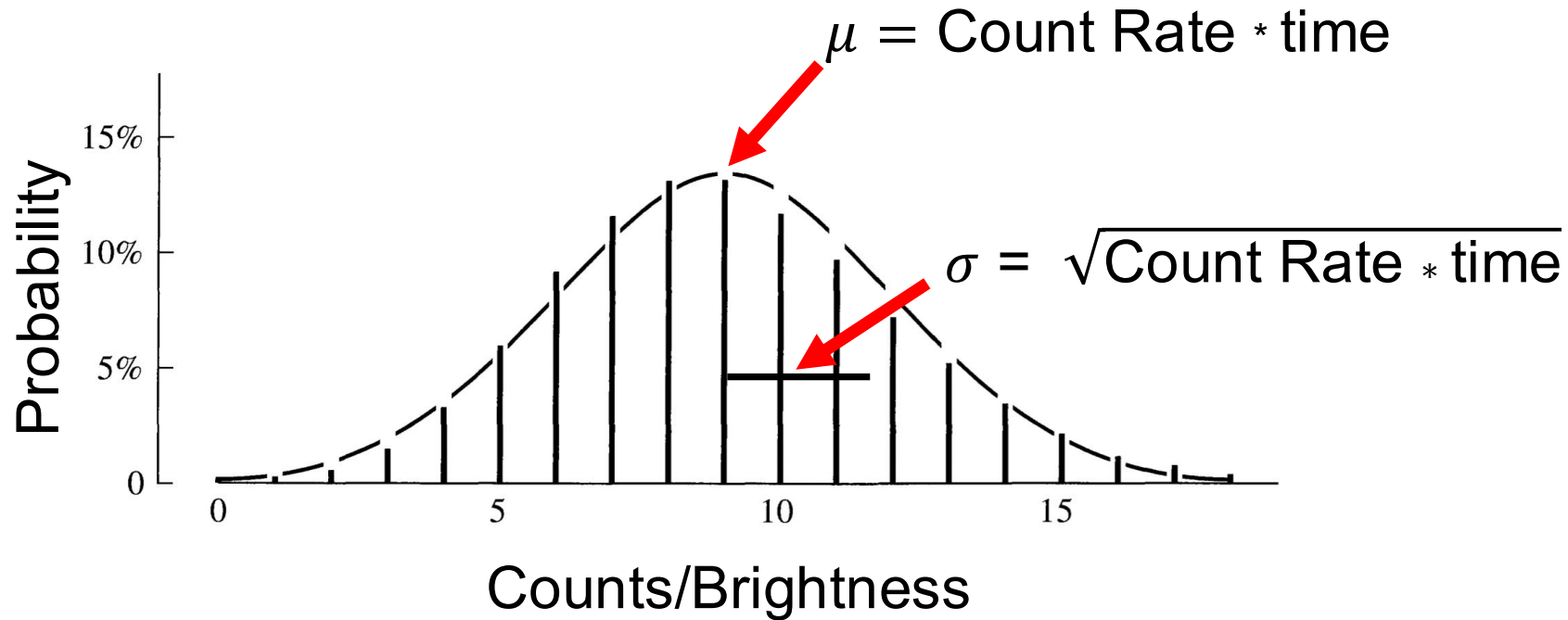
Photon Noise Limit Application: Bright Objects



Expected Value or Mean (μ) = Count Rate * time

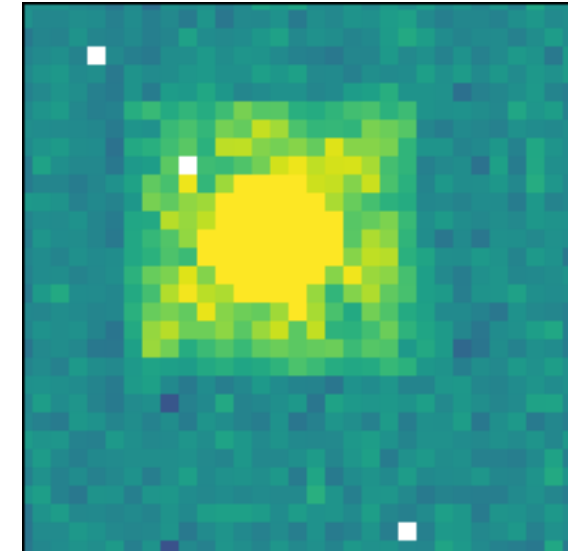
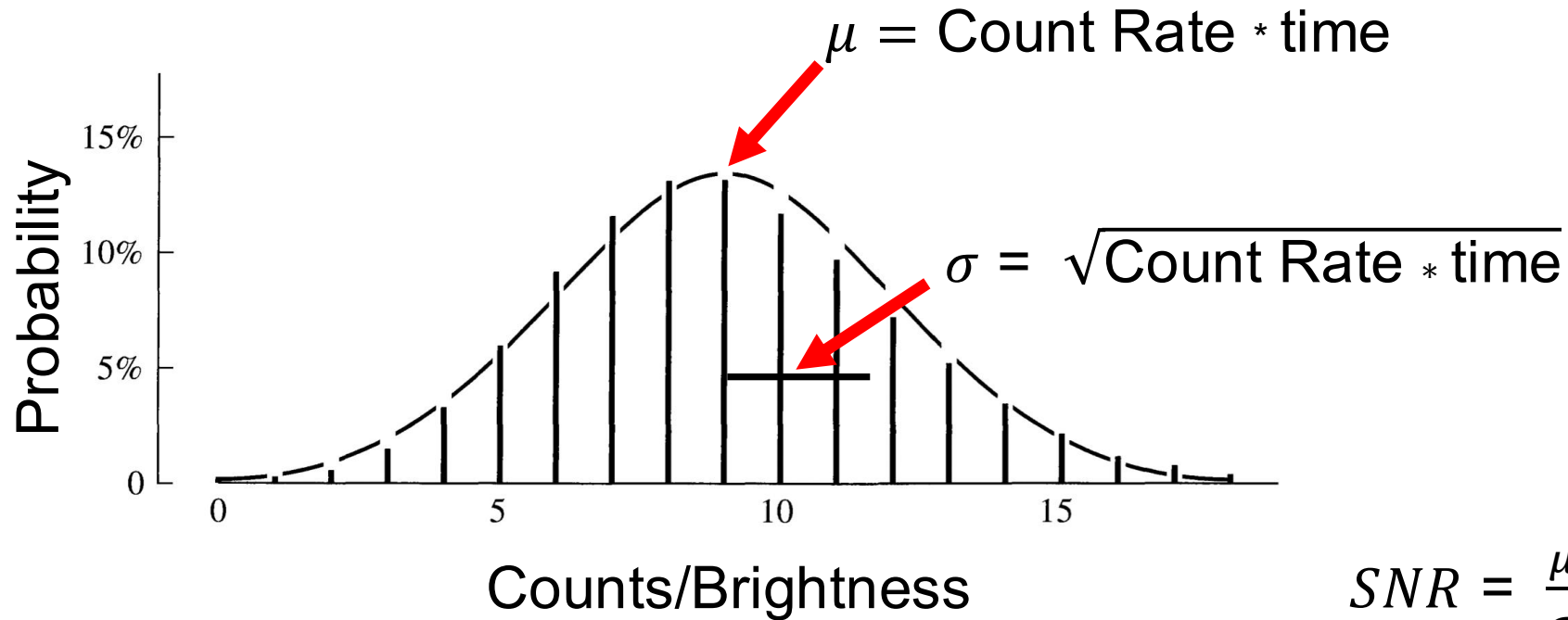
Standard Deviation or Error (σ) = $\sqrt{\text{Count Rate} * \text{time}}$

Photon Noise Limit Application: Bright Objects



If you observe for a longer time the mean value gets bigger, but the standard deviation gets bigger at a slower rate. More time leads to a more precise measurement.

Photon Noise Limit Application: Bright Objects



$$SNR = \frac{\mu}{\sigma} = \sqrt{\text{Count Rate} * \text{time}}$$

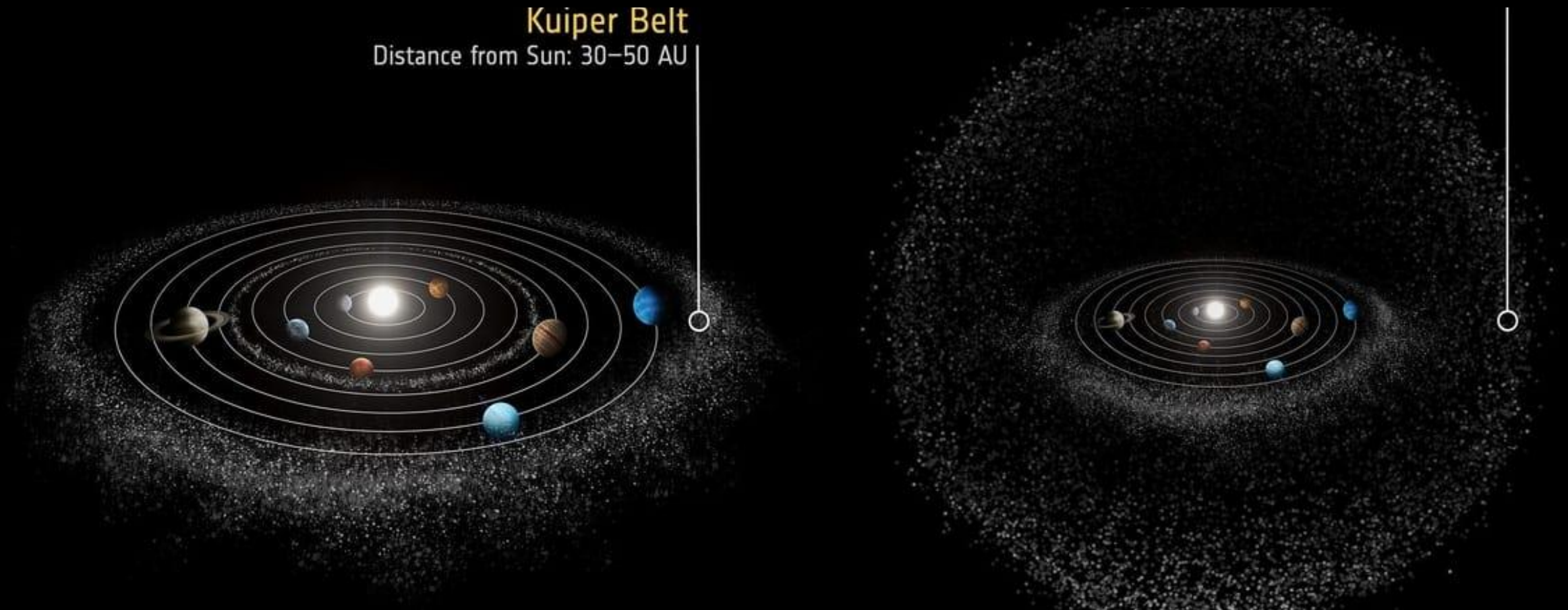
Telescope time is precious. Just because you can get an ultra precise measurement doesn't mean you should. You need to estimate how much time is necessary for the question you are posing.

Coordinates

Where Things Are **Located** is Just as Important as What They Are

Astrometry is the science of measuring the positions and motions of celestial objects.

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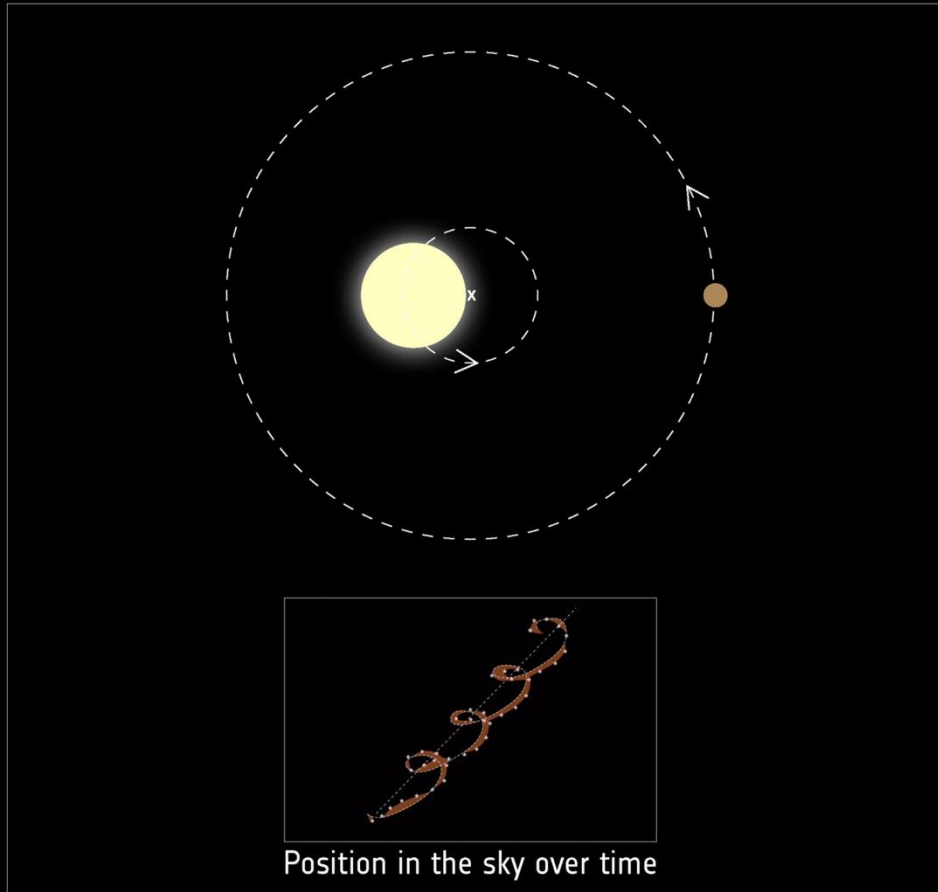
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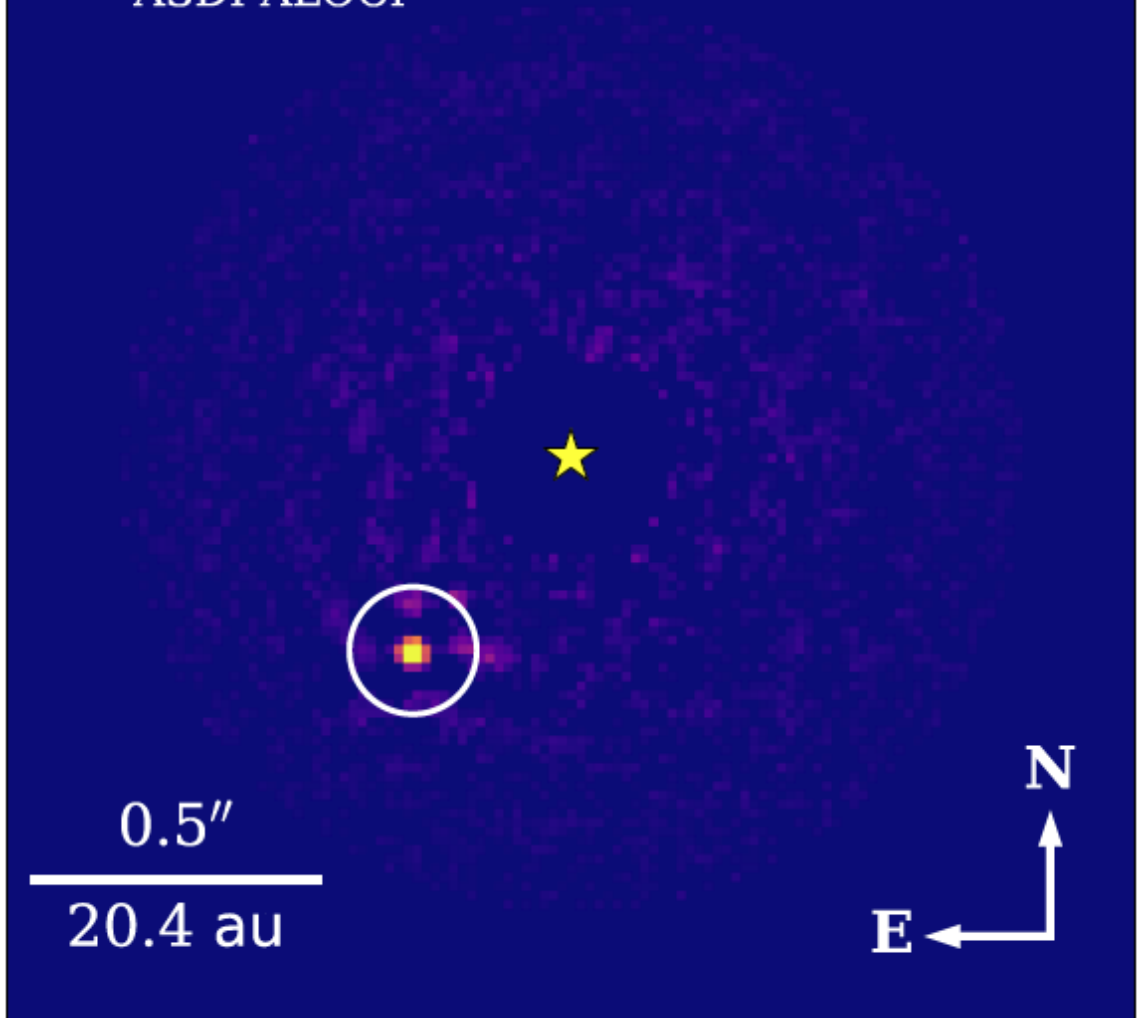
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Astrometry



SCEXAO/CHARIS 19 May 2021
ASDI-ALOC1

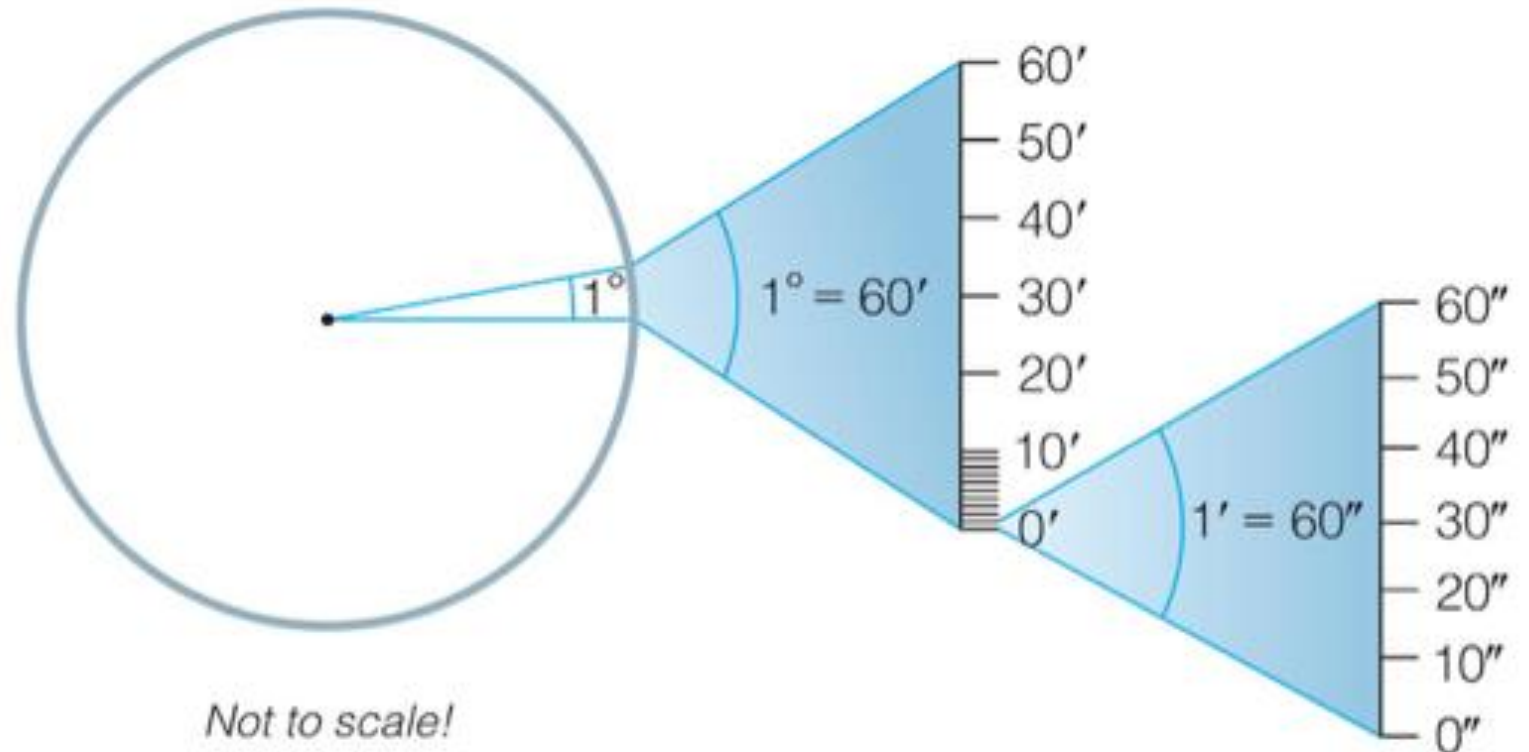


If you have enough location precision, you can infer the presence of an exoplanet around another star with astrometry.

We Describe Positions on the Sky in Angles

Angular Measurements

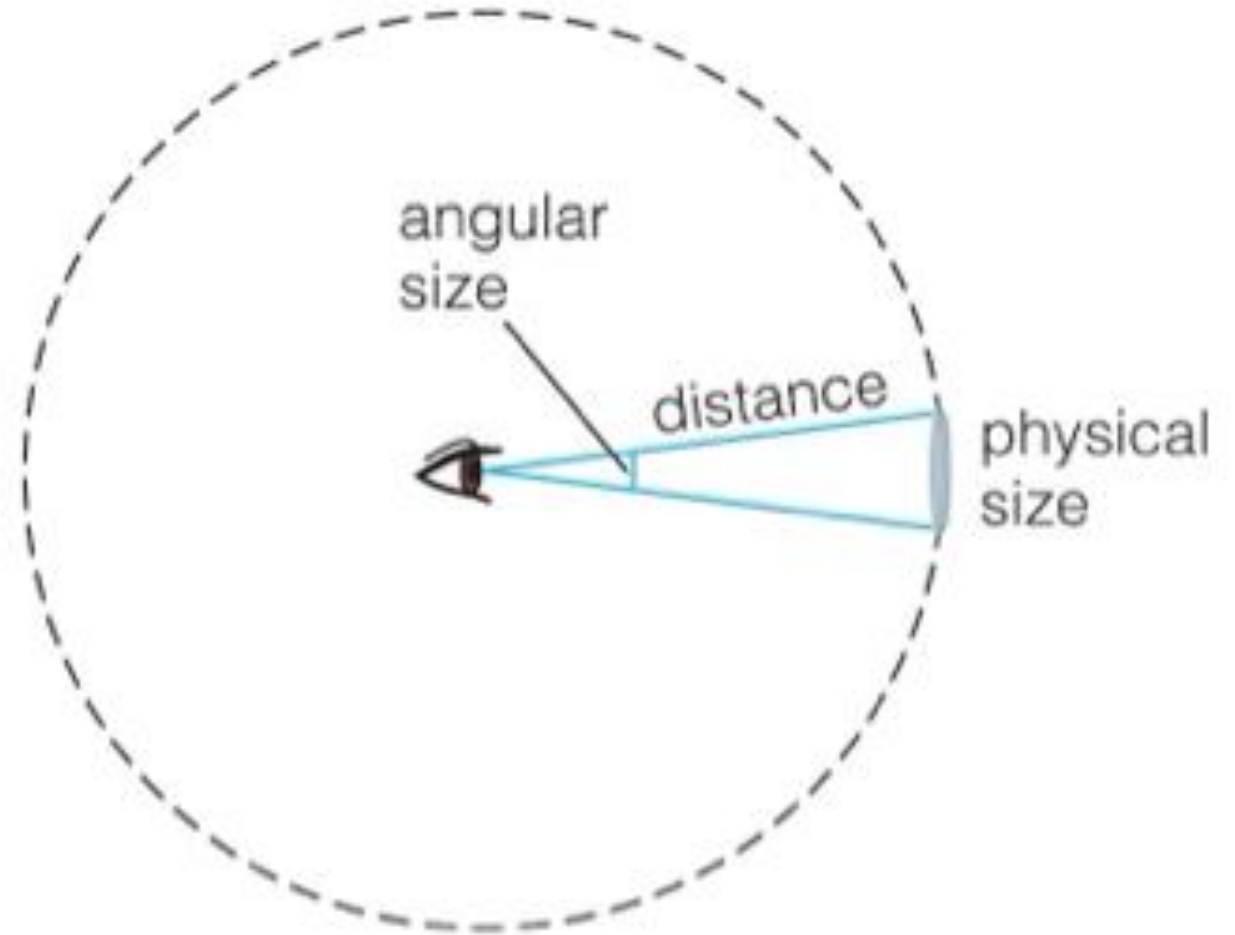
- Full circle = 360°
- $1^\circ = 60'$ (arcminutes)
- $1' = 60''$ (arcseconds)
- 1 hour = 15 degrees



Angular Size

$$\text{angular size} = \text{physical size} \times \frac{360 \text{ degrees}}{2\pi \times \text{distance}}$$

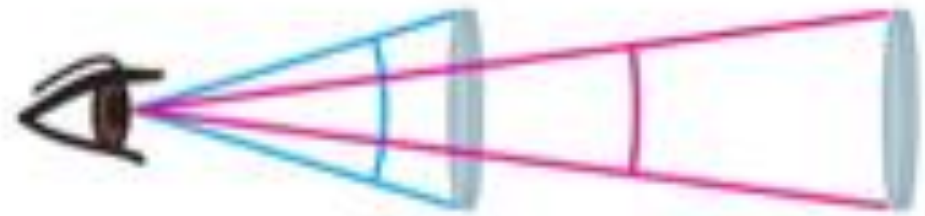
Angular size is related to distance and size.



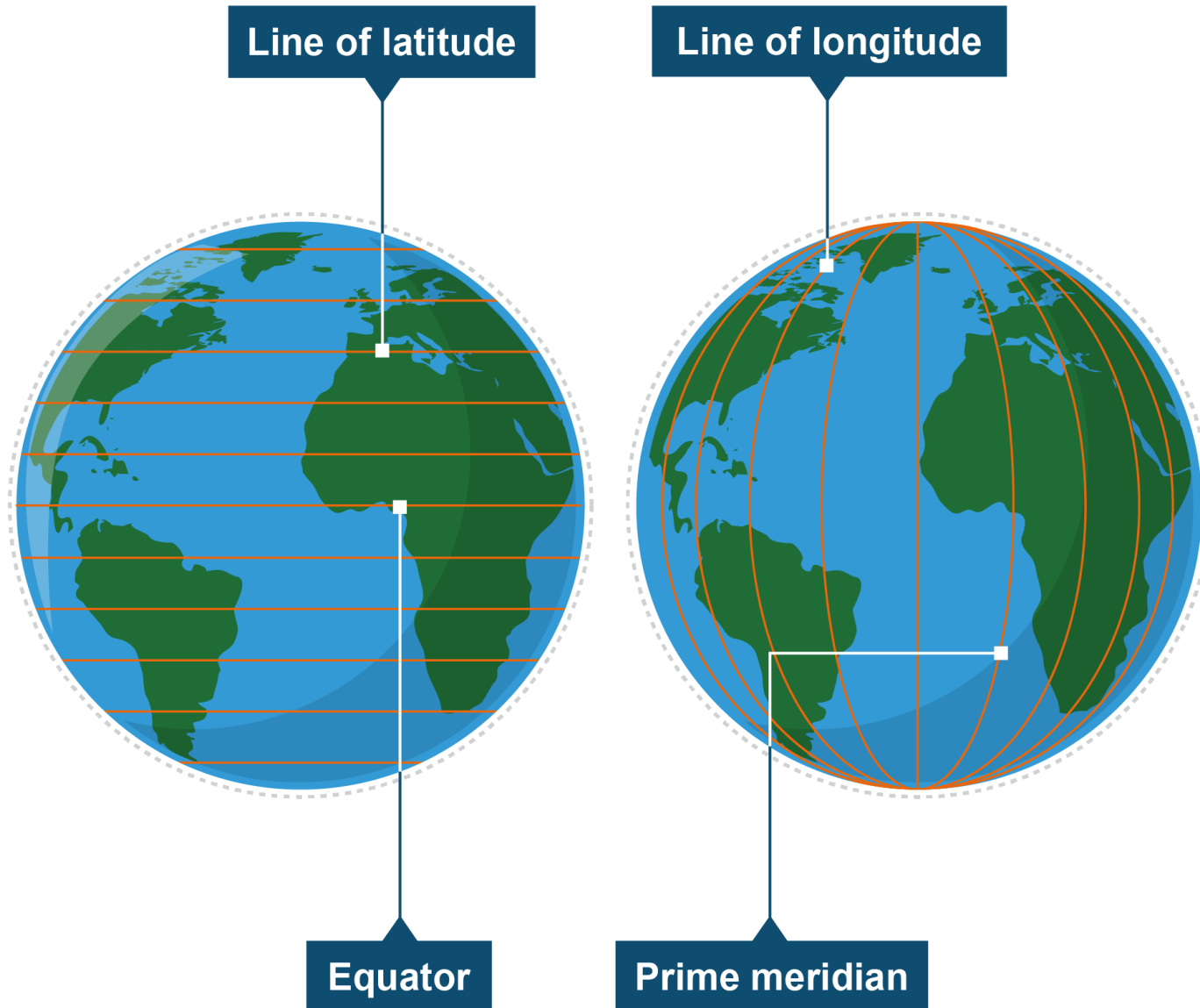
Angular Size

$$\text{angular size} = \text{physical size} \times \frac{360 \text{ degrees}}{2\pi \times \text{distance}}$$

Angular size is related to distance and size. If an object is further away it has a smaller angular size on the sky.



Longitude and Latitude on a Sphere



Latitude – position north or south of equator.

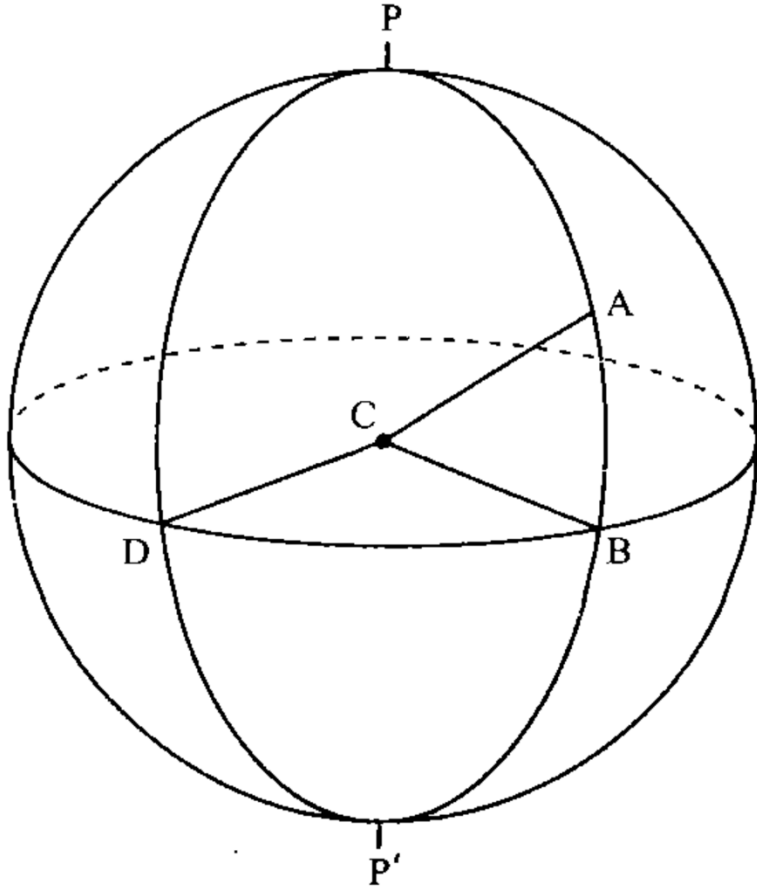
Longitude – position east or west of the prime meridian.

You need to pick an arbitrary longitude line to make a relative measurement of longitude.

Prime Meridian – Longitude = 0

Longitude and Latitude on a Sphere

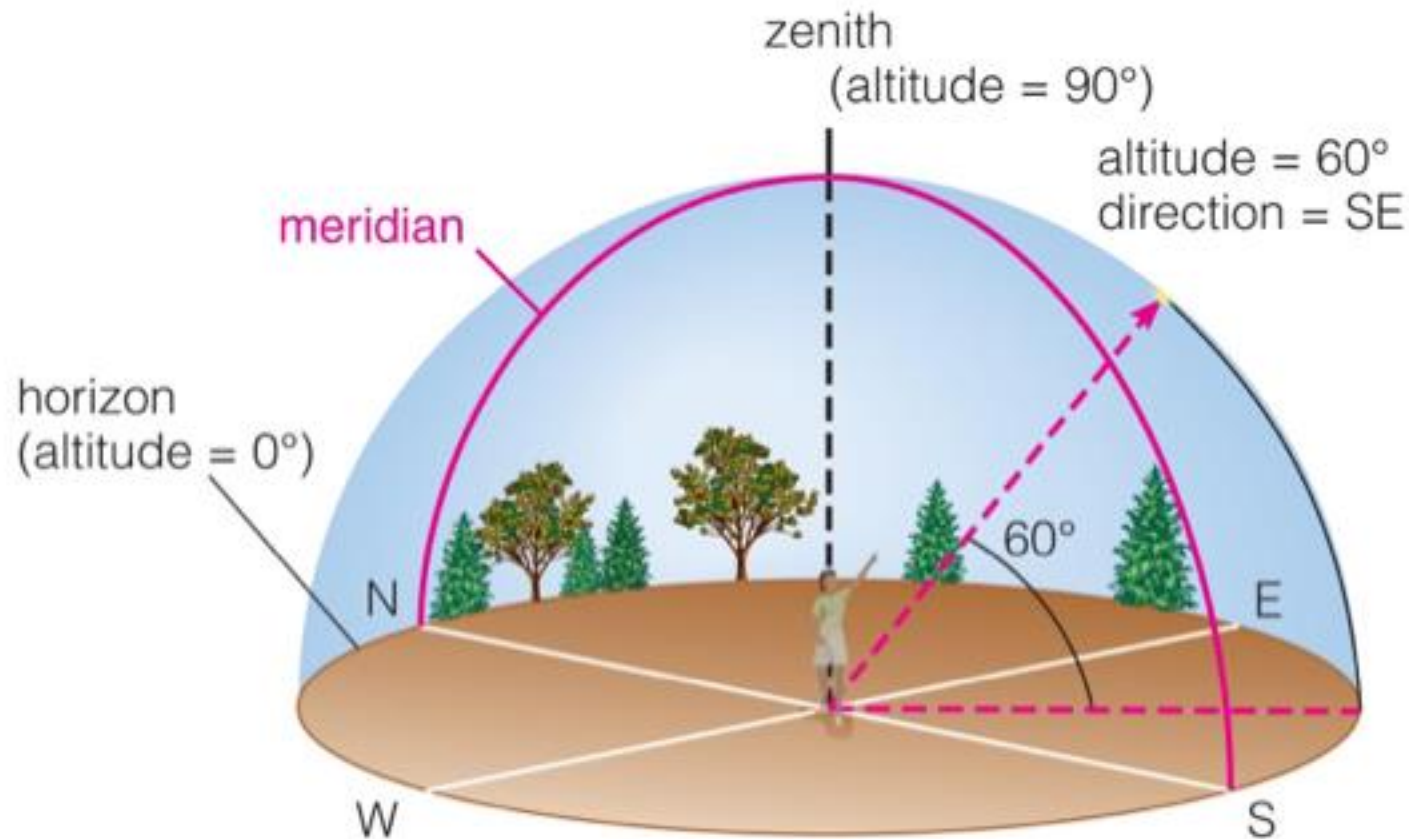
(c)



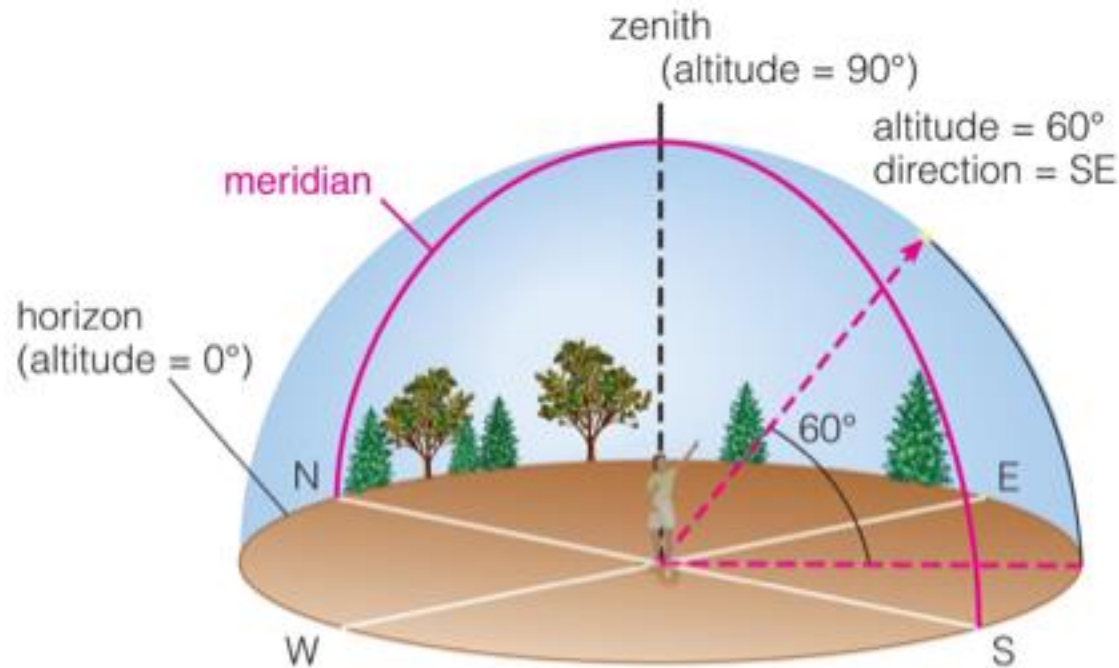
You only need two angles to locate something on a sphere.

The Local Sky: Altitude - Azimuth System

An object's **altitude** (above horizon) and **azimuth** (direction along horizon) specifies its location in your local sky



The Local Sky: Altitude - Azimuth System

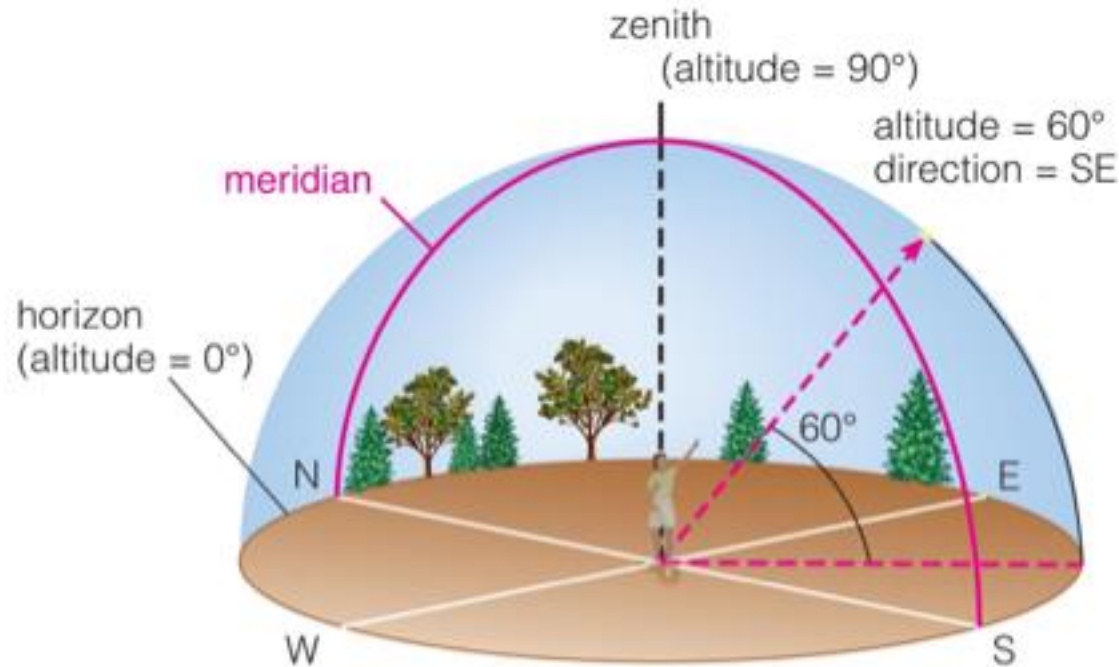


Zenith: The point directly overhead

Horizon: All points 90° away from zenith

Meridian: Line passing through zenith and connecting N and S points on horizon

The Local Sky: Altitude - Azimuth System

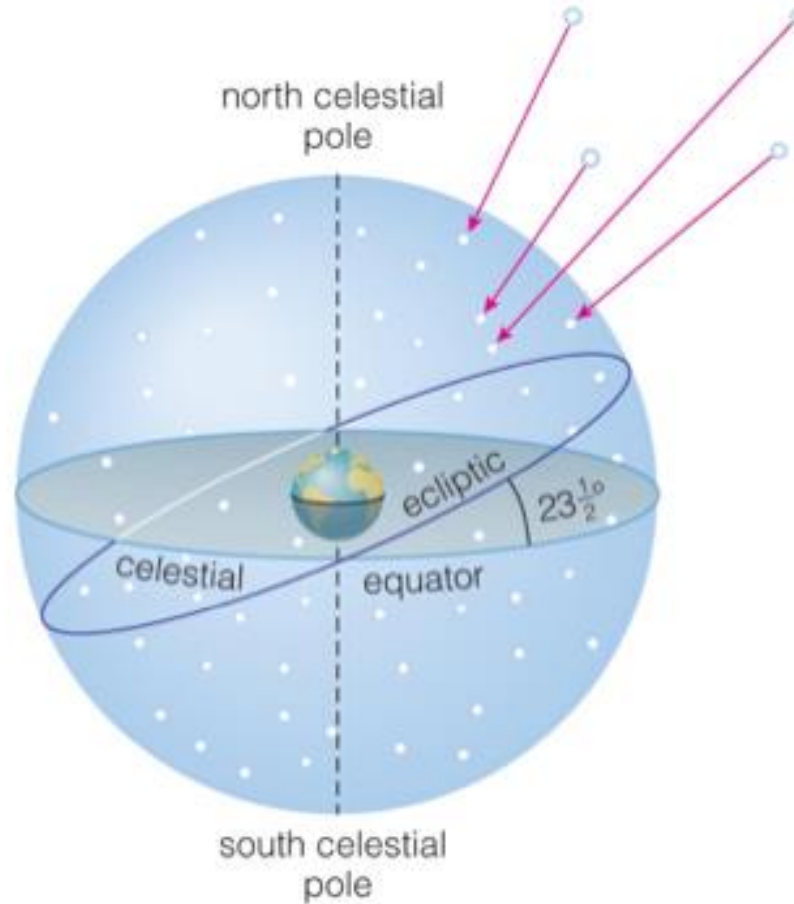


The alt-az system is very intuitive, but making the horizon your reference causes issues.

The coordinates are not “fixed” because Earth rotates towards the East.

In the alt-az system coordinates change over time.

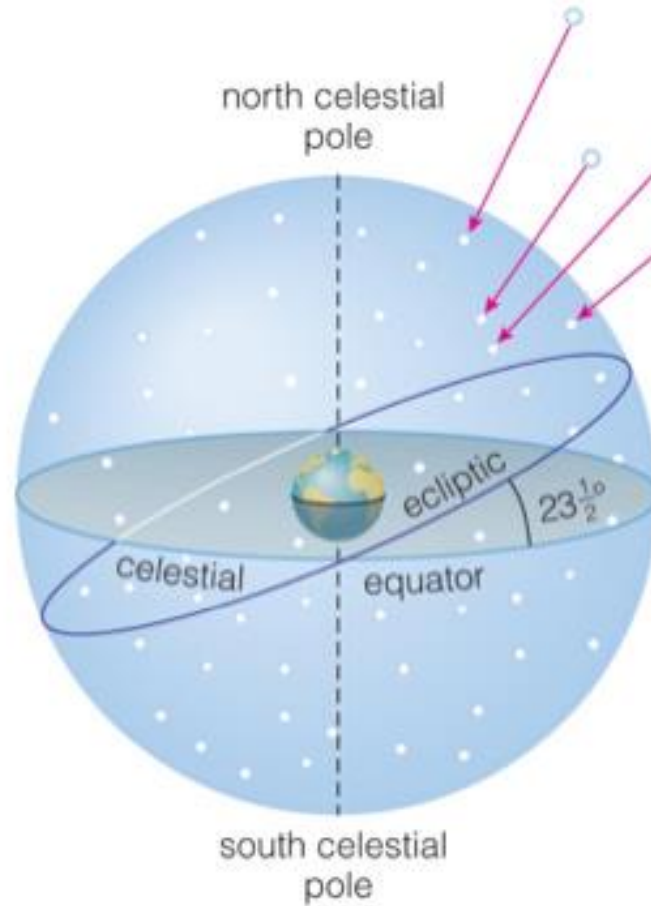
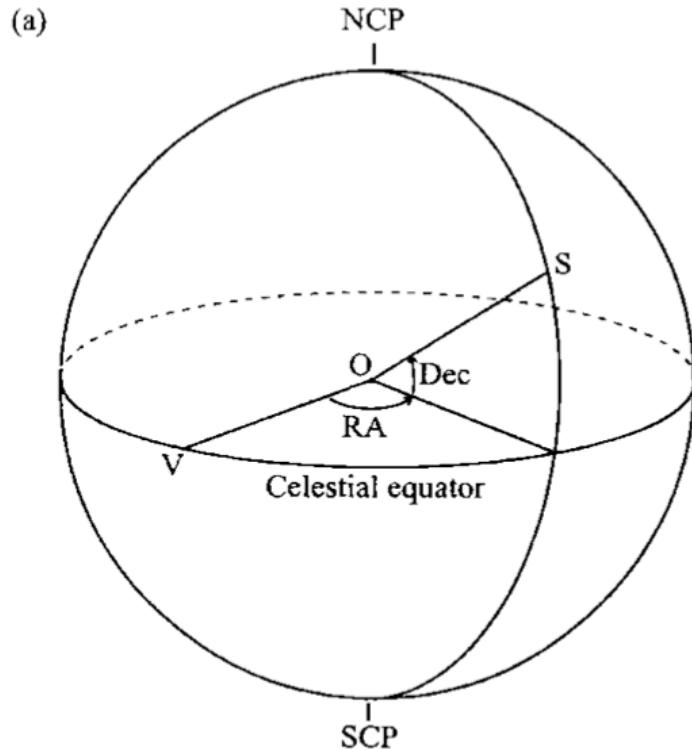
The Celestial Sphere and Equatorial Coordinates



Stars at different distances all appear to lie on the celestial sphere.

Ecliptic is Sun's apparent path through the celestial sphere.

The Celestial Sphere and Equatorial Coordinates

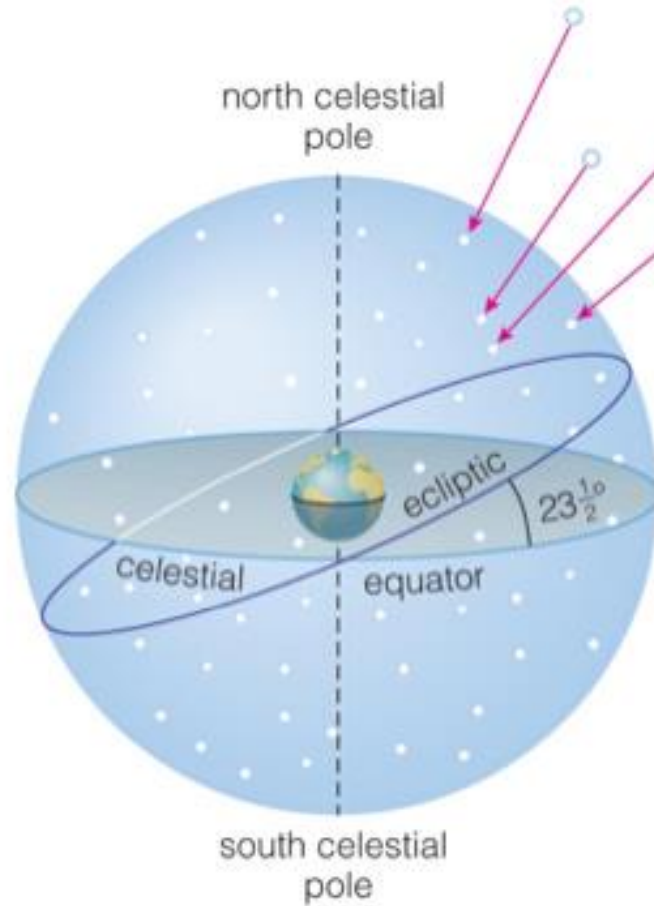
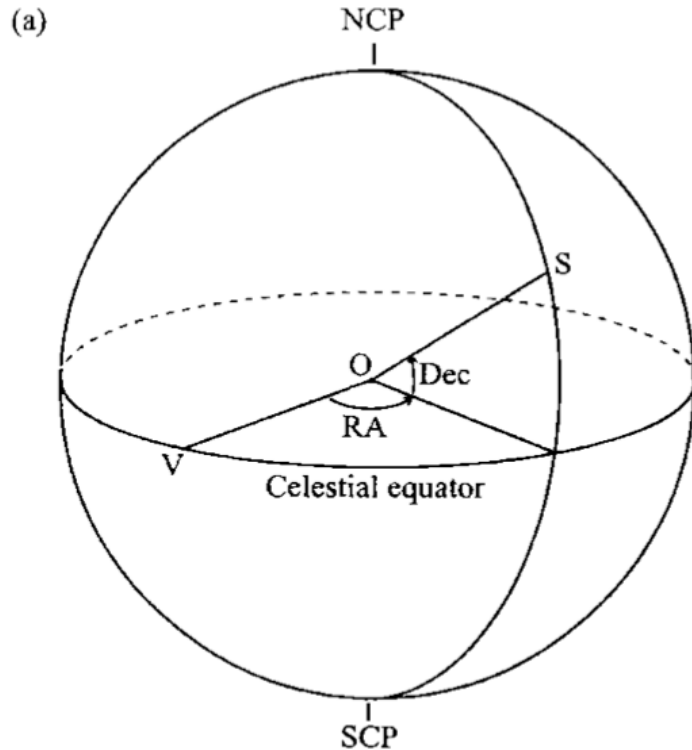


In the equatorial coordinate system, you quote two angles: **Right Ascension (RA)** and **Declination (DEC)**

The relative right ascension value is dependent on where the ecliptic crosses the celestial equator.

RA and **DEC** values are typically quoted with an equinox year and epoch year.

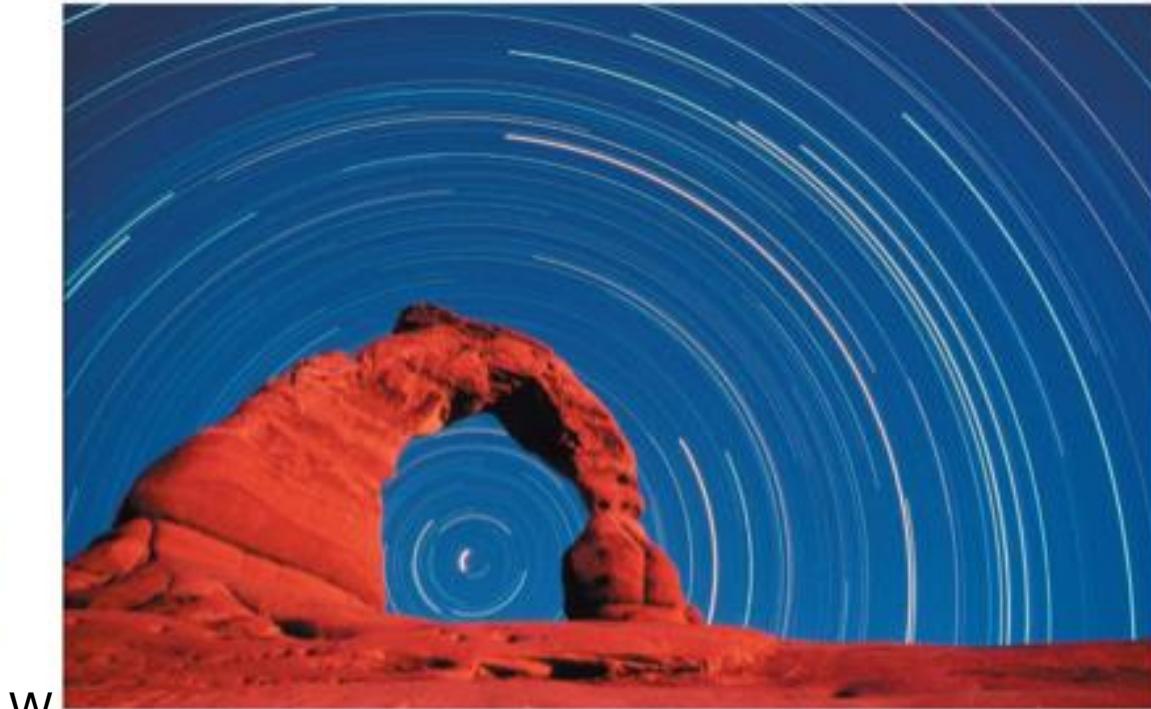
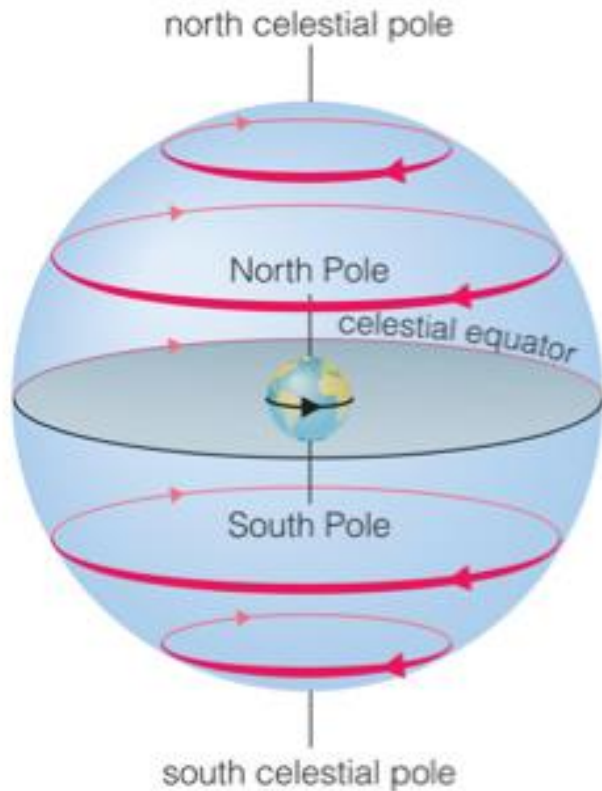
The Celestial Sphere and Equatorial Coordinates



Because the equatorial coordinate system is defined at a specific time it is “fixed” in RA and DEC.

Unlike the Alt-Az system where a target’s angles constantly change over time.

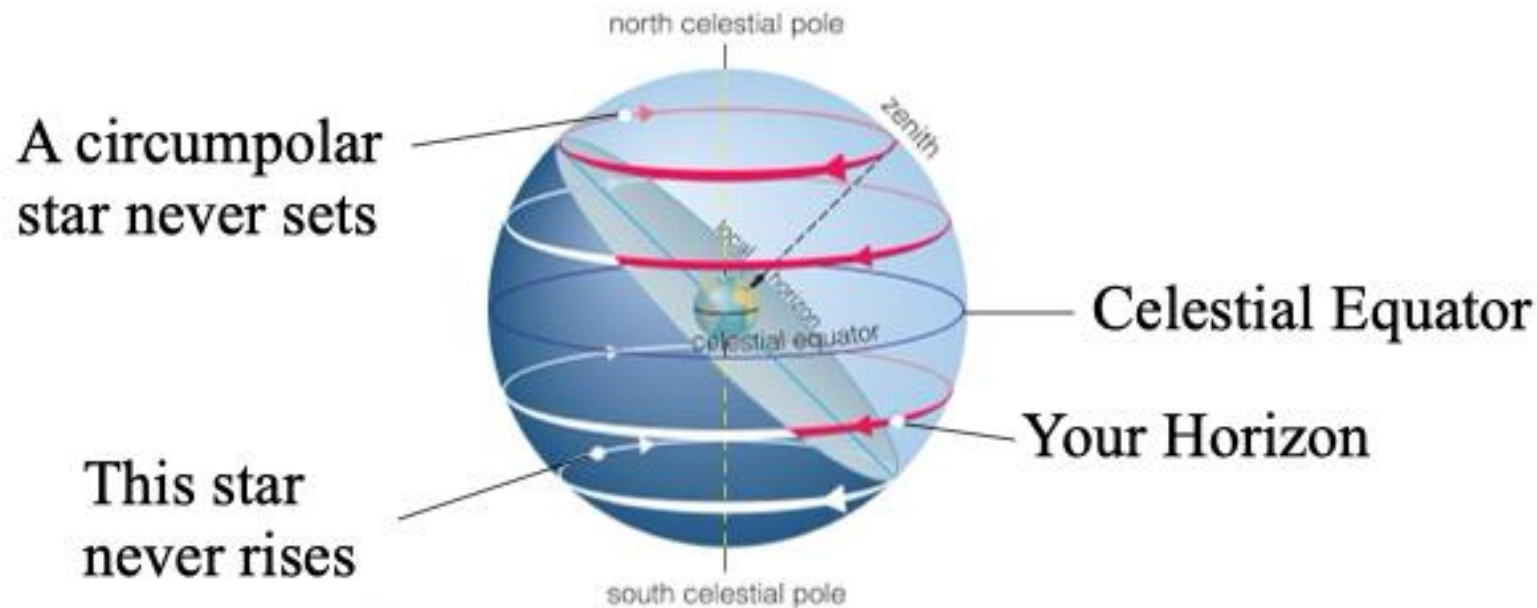
Stars Rise and Set in a Circle



Earth rotates west to east, so stars appear to circle from east to west.

Our view from Earth:

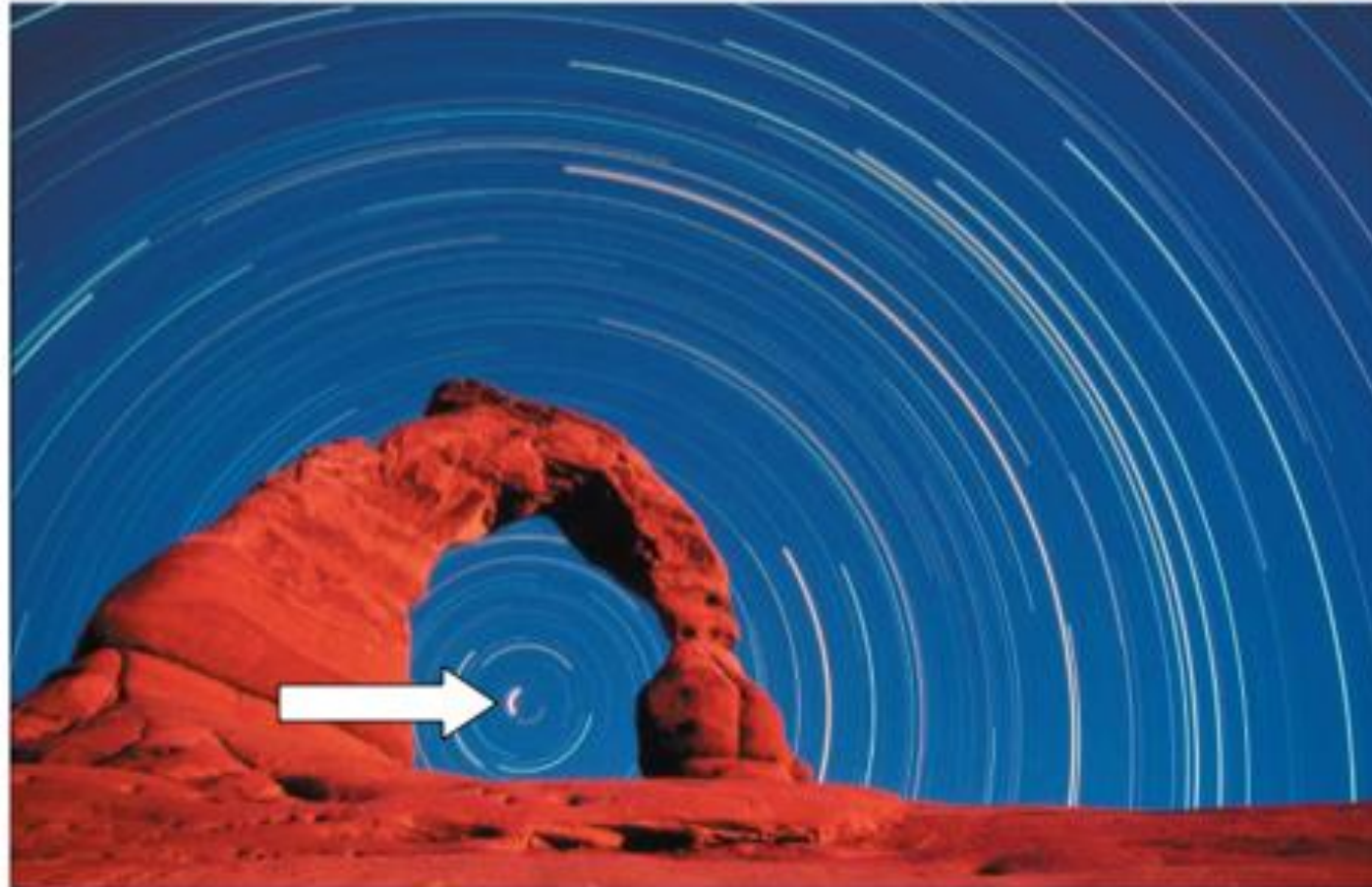
- Stars near the north celestial pole are circumpolar and never set.
- We cannot see stars near the south celestial pole.
- All other stars (and Sun, Moon, planets) rise in east and set in west.



Concept Check

What is the arrow pointing to?

- A. the zenith
- B. the north celestial pole
- C. the celestial equator

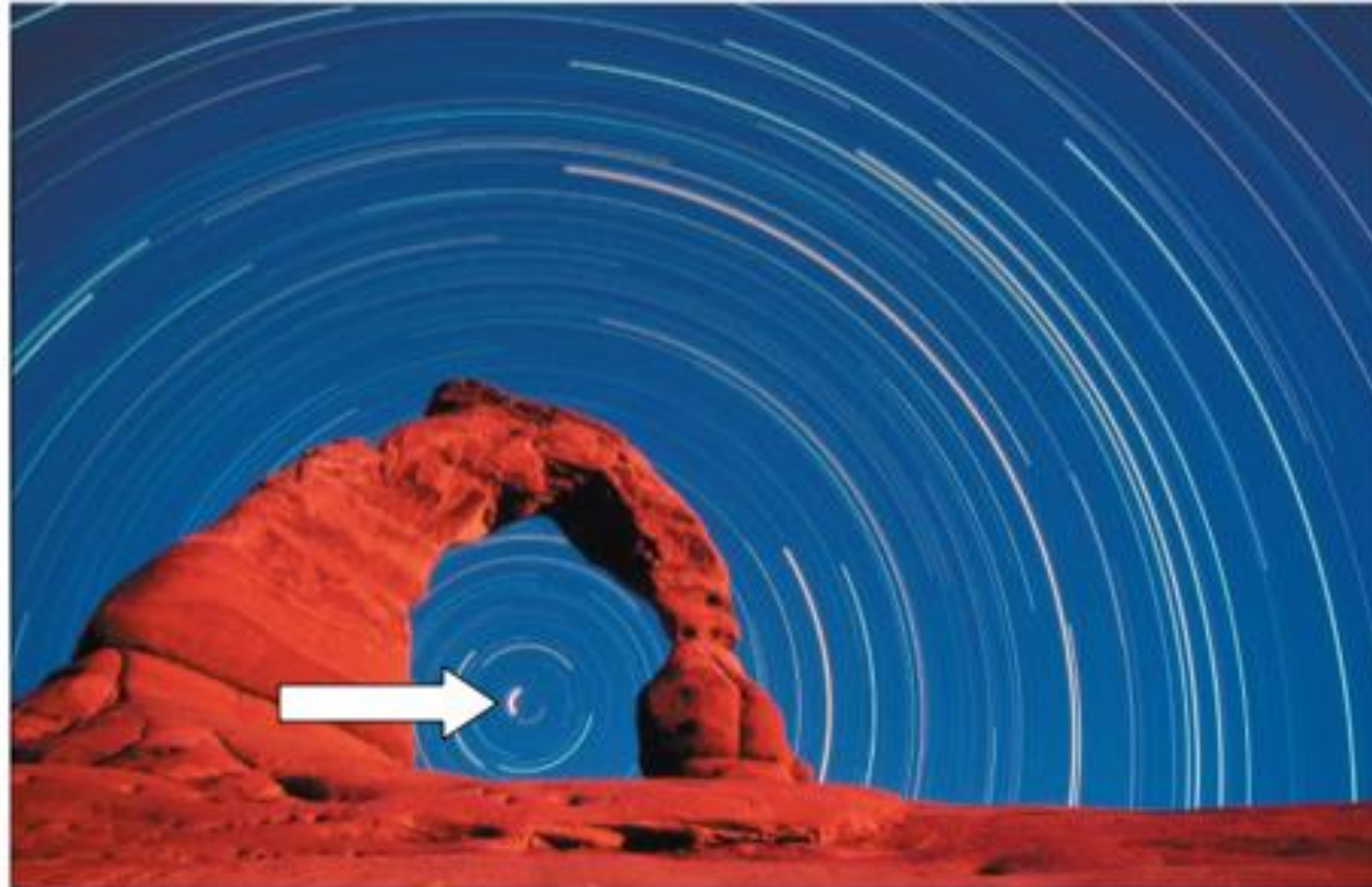


Concept Check

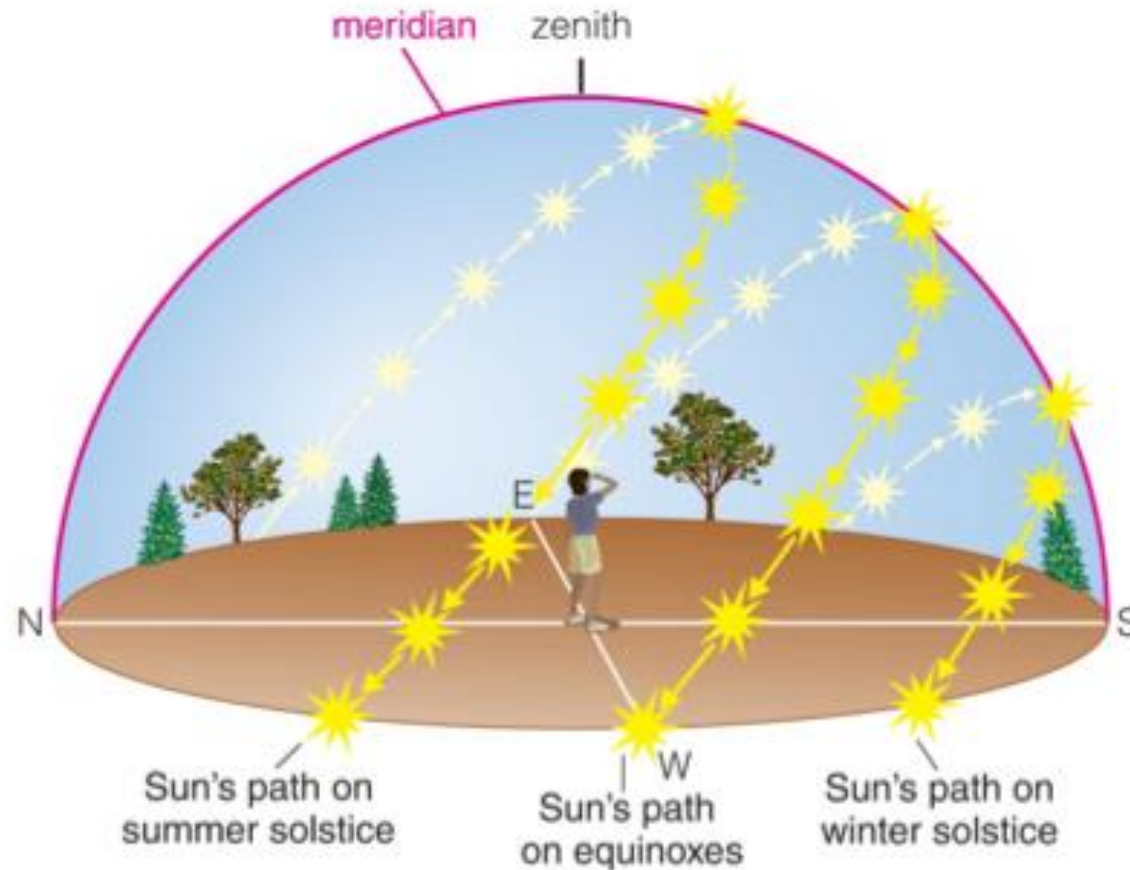
What is the arrow pointing to?

- A. the zenith
- B. the north celestial pole
- C. the celestial equator

B. Your latitude determines the height of the north or south celestial pole



Seasonal/Temporal Changes: The Sun

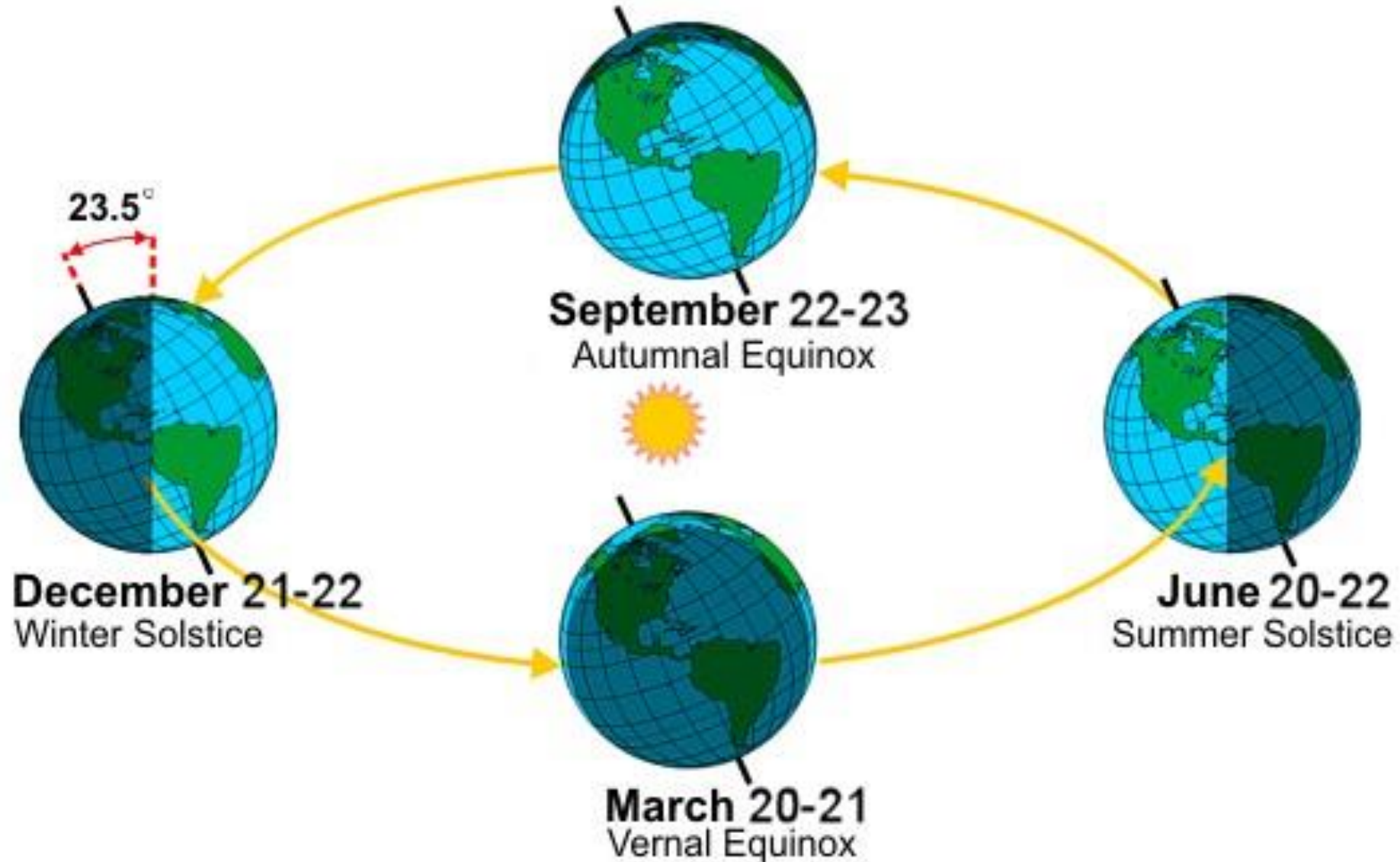


Summer solstice: Highest path, rise and set at most extreme north of due east.

Winter solstice: Lowest path, rise and set at most extreme south of due east.

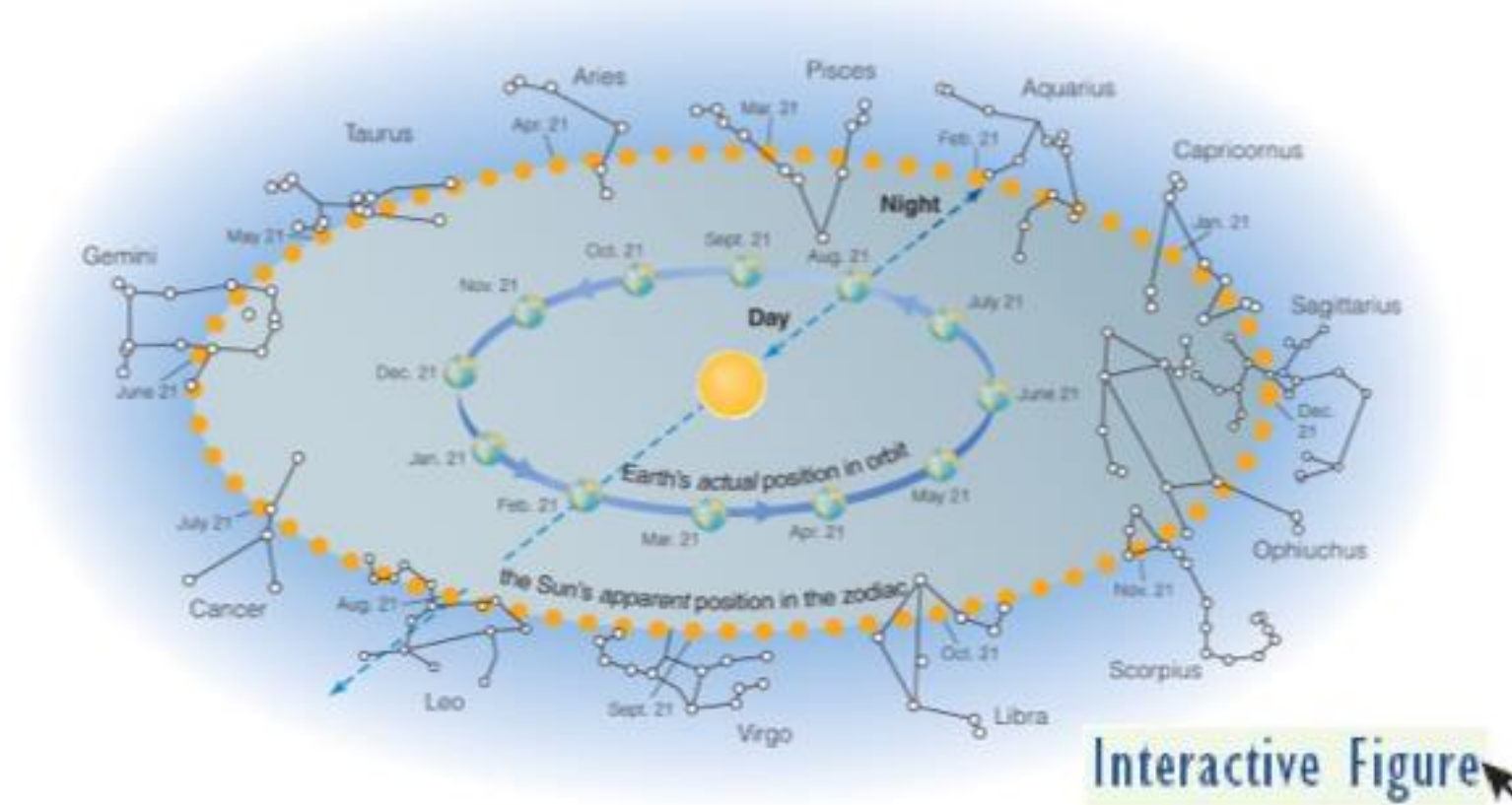
Equinoxes: Sun rises precisely due east and sets precisely due west.

Seasonal / Temporal Changes: The Sun



Seasonal / Temporal Changes: Constellations

- As the Earth orbits the Sun, the Sun appears to move eastward along the ecliptic.
- At midnight, the stars on our meridian are opposite the Sun in the sky.
- **The parts of the sky you can observe change with the seasons.**



The Milky Way Galaxy

